



This is a planning mystery! We know the tasks, the time they take, and the finish line. Can we build a plan?

MISSION: Beat the Clock

The problem

It is 4:10 p.m. Amira needs 20 minutes to finish homework and 15 minutes to clean up. Can she finish both before 5:00 p.m.?

CLUE 1 - What is the problem asking you to find?

Say it in your own words. No calculating yet.

CLUE 2 - What do you KNOW about the plan?

Write the starting time, the deadline, and how long each task takes. Which facts might help you test a plan?

YOUR FIRST PATH - What will you try?

Make a timeline, combine the task times, jump forward, or work backward from 5:00. Pick a path to test.

THINKING TOKENS



I found the facts



I made a plan



I checked the plan



A plan is just an idea we can test. If the first path feels messy, another path may be easier!

Use hints one at a time.

Tip for printing: cover each hint with a sticky note or a strip of scrap paper. Reveal only the next one you need.

HINT 1 - FIND THE FINISH LINE

What time must everything be done? How much time is there from 4:10 to that finish line?

HINT 2 - LOOK AT THE TASKS

There are two jobs. How long would both jobs take if you did one and then the other?

HINT 3 - BUILD A TIMELINE

Start at 4:10. Where are you after 20 minutes? Then move 15 more minutes.

HINT 4 - CHECK THE FINISH

Compare the time your plan ends with 5:00. Does it finish before, at, or after the deadline?

BACK TO YOUR PATH

Now try again. What did the hint help you notice?



You built a plan and tested it! Show your path first. Then tell me whether the plan works.

1. Show your path

Show how you tested whether both tasks fit before 5:00. A timeline, jumps, or another plan all count.

2. Your answer

Can she finish both before 5:00 p.m.?

3. Another door?

Can you test the same plan a different way - for example, by comparing total task time with available time or by working backward?

My second idea:

4. Choose the Thinking Moves you used

- | | |
|--|---|
| ☐ Found the deadline | ☐ Sequenced the tasks |
| ☐ Tested a plan | ☐ Checked the finish |

Purpose of this mission

This mission practices sequencing and feasibility checking. The arithmetic is intentionally light. The goal is for Amira to turn a real situation into a plan, test whether the plan fits a constraint, and recognize that there are several valid ways to check the same plan.

What to praise

- "You found the deadline before calculating."
- "You turned the story into a plan you could test."
- "You checked where your plan actually ended."
- "You tried another way to see if the plan still worked."

Answer

Yes. If she starts at 4:10, 20 minutes of homework ends at 4:30, and 15 minutes of cleanup ends at 4:45. She finishes before 5:00.

Valid reasoning paths to recognize

Build the timeline

$4:10 + 20 \text{ minutes} = 4:30$. Then $4:30 + 15 \text{ minutes} = 4:45$. That is before 5:00.

Compare time needed with time available

The two tasks need 35 minutes total. From 4:10 to 5:00 there are 50 minutes, so 35 minutes fits.

Work backward from the deadline

To finish a 15-minute cleanup by 5:00, cleanup could start by 4:45. A 20-minute homework task before that could start by 4:25. Starting at 4:10 is early enough.

If she gets stuck

Ask: "What is the finish line?" Then: "What are the pieces of the plan?" If she has the facts but stalls, ask her to draw a simple timeline from 4:10 to 5:00 and place one task on it. Avoid telling her which operation to use.